Multiple Choice Questions

1 point MCQs

Question 1 (1 point)

What is the regular expression for the following language
$L = \{w \in \{a, b\}^* \mid w$ has no two consecutive a's or b's and has at least one $a\}$?

A. $(b + \epsilon)a(ba)^*(b + \epsilon)$

B. $(a + \epsilon)b(ab)^*(a + \epsilon)$

C. $(ba)^*$

D. $ba(ba)^*(b + \epsilon)$

Question 2 (1 point)

If $L = \{0, 10, 11\}$, , which of the following is in L^*?

A. 01001.

B. 100011.

C. 11111.

D. 0010111.

Question 3 (1 point)

How many strings of length at most 3 is contained in the language described by the following regular expression, $(x + y)^*y(a + ab)^*$?

A. 7.

B. 10.

C. 11.

D. 12.

Question 4 (1 point)

Which of the options is equivalent to the following regular expression : $\epsilon + 1^*(011)^*(1^*(011)^*)^*$.

A. $(1^*(011)^*)$

B. $(1 + (011)^*)^*$

C. $(1 + 011)^*$

D. $(1011)^*$

Question 5 (1 point)

Which of the following statements are correct?

A. There is a 2^k-state DFA for every k-state NFA.

B. There is a k-state DFA for every 2^k-state NFA.

C. There is a k-state NFA for every 2^k-state DFA.

D. There is a 2^k-state NFA for every k-state DFA.

Question 6 (1 point)

Which of the following regular expression are equivalent?

$R_1 : a^*(ab^* + b^*) + b^*(ba^* + a^*)$
$R_2 : (a^* + b^*)(a^* + b^*)$
$R_3 : a^* + bb^*a^*$
$R_4 : a^*b^* + b^*a^*$

A. Only R_2 and R_4.

B. Only R_2, R_3 and R_4.

C. Only R_1, R_2 and R_3.

D. Only R_1, R_2 and R_4.

Question 7 (1 point)

How many strings of length exactly three does the language described by the following regular expression $(0+1)^*0(0+01)^*$ contains?

A. 5

B. 6

C. 7

D. 8

Question 8 (1 point)

Let P, Q and R be regular expressions such that the number of strings generated by P is p, Q is q and R is r. What is the number of strings generated by the regular expression $(P+R)^*Q+PQ$?

A. $(p+r)q+pq$

B. $2^{(p+r)}+pq$

C. $(2^{(pq)}+q)(p+q)$

D. Infinite